Practical - 8

[page Replacement Policy]

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

INPUT :

```
~
  GNU nano 8.7                                                                                                    lru.c
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, f, i, j, k, hit = 0, fault = 0;
    int last_used[10], time = 0, lru_index;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i = 0; i < f; i++) {
        frames[i] = -1;
        last_used[i] = 0;
    }

    for(i = 0; i < n; i++) {
        int found = 0;

        for(j = 0; j < f; j++) {
            if(frames[j] == pages[i]) {
                hit++;
                time++;
                last_used[j] = time;
                found = 1;
                break;
            }
        }

        if(found == 0) {
            int min = last_used[0];
            lru_index = 0;

            for(j = 1; j < f; j++) {
                if(last_used[j] < min) {
                    min = last_used[j];
                    lru_index = j;
                }
            }

            frames[lru_index] = pages[i];
            time++;
            last_used[lru_index] = time;
            fault++;
        }
    }

    printf("\nPage Hits = %d", hit);
    printf("\nPage Faults = %d", fault);

    printf("\nHit Percentage = %.2f%%", (float)hit/n*100);
    printf("\nFault Percentage = %.2f%%\n", (float)fault/n*100);

    return 0;
}
```

OUTPUT :

```
~

HP@Hiteshri MSYS ~
$ nano lru.c

HP@Hiteshri MSYS ~
$ gcc lru.c -o lru

HP@Hiteshri MSYS ~
$ ./lru
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Enter number of frames: 3

Page Hits = 2
Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%
```